

SHOPSY ONLINE E-COMMERCE DATABASE MANAGEMENT SYSTEM

CHAPTER-1

1.1INTRODUCTION:

Shopsy is an online e-commerce platform that allows customers to purchase a wide range of products such as fashion, electronics, beauty products, home essentials, and accessories. It provides users with a convenient shopping experience by offering features like product browsing, secure online payments, order tracking, and doorstep delivery.

The Shopsy Online E-Commerce Database Management System is designed to manage important information such as customer details, products, sellers, orders, payments, and inventory in a centralized database. The system helps in storing, retrieving, updating, and managing data efficiently while reducing redundancy and maintaining data accuracy. It also supports smooth business operations and improves customer satisfaction by ensuring quick access to information.

1.2PROBLEMSTATEMENT:

Managing customer records, products, orders, payments, and inventory manually can be time-consuming and may lead to errors, duplicate data, and poor data management. As the number of users and products increases, it becomes difficult to maintain accurate records without a proper database system. This project aims to solve these challenges by designing a structured database that stores all business information efficiently. It ensures data consistency, improves security, simplifies order management, and supports faster retrieval of information.

1.3 OBJECTIVE :

The main objective of this project is to design and develop a database for the Shopsy Online E-Commerce platform that efficiently manages customers, products, sellers, orders, payments, and inventory. The database aims to improve data accuracy, reduce redundancy, maintain data integrity, and support smooth business operations through efficient data management.

1.4 KEY OBJECTIVE :

- To design a relational database for the Shopsy platform.
- To manage customer and seller information.
- To maintain product and category details.

- To process customer orders and payments.
- To manage inventory and product availability.
- To establish relationships using Primary Keys and Foreign Keys.
- To reduce data redundancy through normalization.
- To ensure secure and efficient data management.

1.5 BUSINESS REQUIREMENT :

The database should support customer registration, product management, seller management, order processing, payment handling, inventory management, and delivery tracking. It should also maintain customer order history and generate reports for business analysis. The system should provide accurate data storage, quick retrieval of information, and secure management of customer and business records.

1.6 STAKEHOLDER :

The major stakeholders of the project include Customers, Sellers, Administrators, Delivery Partners, and Business Management. Customers purchase products and place orders, sellers manage products and inventory, administrators maintain the database, delivery partners update delivery status, and management uses reports for business decisions.

1.7 FUNCTIONAL REQUIREMENT :

The system should provide customer registration and login, product management, category management, seller management, shopping cart, order processing, payment management, inventory management, delivery tracking, customer reviews, report generation, and data retrieval. These functions help the platform operate efficiently and provide a better shopping experience.

1.8 NON – FUNCTIONAL REQUIREMENT

The system should provide high performance, security, reliability, scalability, and data integrity. It should retrieve data quickly, protect customer information, and support future expansion. The database should also be easy to maintain and ensure regular backup and recovery of data.

1.9 MOTIVATION OF THE PROJECT :

The motivation behind this project is to understand how a real-world e-commerce platform manages large amounts of data using a Database Management System. It provides practical knowledge of database design, SQL, normalization, and table relationships. This project also helps improve database management skills while

demonstrating how an efficient database supports smooth business operations, reduces manual work, and enhances the overall customer shopping experience. This version is college-project standard—not too short, not too lengthy, and should fit well in a Word document. If you're also making a mini project report (30–40 pages), this length is appropriate.